WILLIAM O. HOLDEN

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SUMMARY

AI Engineer building internal agentic and retrieval-augmented applications at Aderant for the teams behind the company's legal software. Background spans full-stack development, LLM orchestration, and deep learning for computer vision and NLP, with peer-reviewed research in *Computers and Electronics in Agriculture*.

PROFESSIONAL EXPERIENCE

Aderant Dec 2025 – Present
AI Engineer, Docketing & VI Engineering *Atlanta, GA (Hybrid)*

- Build internal AI applications for the teams behind Aderant's legal software, focused on agentic workflows and retrieval-augmented generation.
- Stack: Python, TypeScript/React, AWS, Claude Platform, Terraform.

University of Georgia, School of Computing 2024 – Dec 2025
Graduate Research Assistant, Computer Vision *Athens, GA*

- Built a spatiotemporal video encoder with metadata fusion (X3D 3D-CNN + YOLO tracking) for automated broiler-chicken behavior recognition from smartphone farm footage: 86% test accuracy (macro F1 0.86) across five behaviors.
- Developed motion-correction algorithms to normalize avian kinematics and remove human-handling bias, enabling reliable velocity and acceleration metrics from handheld video.
- 1st Place, UGA Artificial Intelligence in Livestock and Poultry Farming Conference poster competition (Sep 2025).

SELECTED PROJECTS

DeckTection: Trading Card Identification Pipeline PyTorch, YOLOv12, SAM2

- Engineered an end-to-end CV pipeline (YOLOv12 → SAM2 → Siamese network); fine-tuned YOLOv12 to 99.49% mAP50 and trained a ResNet-18 Siamese network to 95% one-shot accuracy across 4,000 cards.

Whisper Inference Optimization ONNX Runtime, Quantization

- Quantized the Whisper encoder to INT8, cutting model size 70% with no material word-accuracy loss (LibriSpeech); implemented Winograd convolution for 1D layers for 60% compression enabling mobile deployment.

Evaluating LLMs as Translators MATEO, COMET, BLEU

- Benchmarked 15 Korean→English systems (DeepL, Google, GPT-5, Claude, DeepSeek, local Llama/Qwen/Gemma models) against the published human translation; DeepL led COMET (86.5) while Claude, DeepSeek, and GPT-5 beat it on BLEU and chrF2.

Cinema E-Booking Platform React, Django, SQL

- Full-stack booking system; led Agile sprints for a 5-engineer team.

TECHNICAL SKILLS

Languages: Python, JavaScript/TypeScript, Java, C++, C, SQL, R
AI & ML: Agentic workflows & LLM orchestration, RAG, vector search, prompt engineering, Claude Platform, Amazon Bedrock, AgentCore, PyTorch, TensorFlow, scikit-learn, OpenCV, YOLO, SAM, ONNX
Cloud & DevOps: AWS (EC2, ECS, EKS, Lambda, S3, Route 53, Aurora/RDS, Bedrock), Terraform, Flux GitOps, Docker, Grafana, Git/GitHub Actions
Data & Web: React, Node.js, Django, REST APIs, Aurora/PostgreSQL, MongoDB, SQL Server, Aha!, Marketo
Tools: Glean, Browserbase, Playwright, Postman

EDUCATION

University of Georgia, M.S. in Artificial Intelligence, GPA 3.73 Dec 2025
University of Georgia, B.S. in Computer Science, Honors Program, Presidential Scholar, Dean's List Dec 2024

PUBLICATIONS

Holden, W.O., Li, G., Agyemang-Duah, E., et al. "Spatiotemporal video encoder integrated with metadata fusion for action recognition and behavior analysis." *Computers and Electronics in Agriculture* 253 (2026), 112159. doi:10.1016/j.compag.2026.112159 (IF 10.3).

Agyemang-Duah, E., Poku, R.A., Adenyo, C., Osei-Amponsah, R., Holden, W.O., et al. "Climate adaptation in poultry: Spatiotemporal behavioral and kinematic responses of chicken genotypes to identical temperature-humidity exposures." *Journal of Veterinary Behavior* 87 (2026), 41–49. doi:10.1016/j.jveb.2026.08.002.